

Xu Shen, incoming PhD. 沈旭

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Education

- 2026 fall– **PhD. Computer Science, Michigan State University**
Advisor: Prof. Jiliang Tanag.
- 2023 – 2026 **M.Sc. Computer Science, Jilin University**
Advisor: Prof. Xin Wang.
- 2019 – 2023 **B.Sc. Computer Science, Ningbo University**
Co-Worker: Prof. Chengbin Peng, Prof Pietro Liò.

Intern

- 2025.06 – 2025.09 **Summer Intern, University of North Carolina at Chapel Hill**
Advisor: Prof. Tianlong Chen.

Research Interests

- 📌 **Trustworthy Deep Learning**, OOD detection, generalization and representation power of GNNs.
- 📌 **Generative Models**, Theory and Applications of Generative Models (LLMs, Agent, Diffusion Model).

Research Publications (* indicates equal contribution)

Published

- Xu Shen**, Zhang Qi, Wang Song, Zhen Tan, Xinyu Zhao, Laura Yao, Tadiparthi Vaishnav, Nourkhiz Mahjoub Hossein, Morad Ehsan, Lee Kwonjoon, and Tianlong Chen, “Metacognitive self-correction for multi-agent system via prototype-guided next-execution reconstruction,” in *Findings of The 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
- Xu Shen**, Wang Song, Zhen Tan, Laura Yao, Xinyu Zhao, Kaidi Xu, xin Wang, and Tianlong Chen, “Faithcot-bench: Benchmarking instance-level faithfulness of chain-of-thought reasoning,” in *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- Xu Shen**, Yixin Liu, Yiwei Dai, Yili Wang, Rui Miao, Yue Tan, Shirui Pan, and Xin Wang, “Understanding the information propagation effects of communication topologies in llm-based multi-agent systems,” in *The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

- 4 Yili Wang*, Yixin Liu*, **Xu Shen***, Chenyu Li, Kaize Ding, Rui Miao, Ying Wang, Shirui Pan, and Xin Wang, "Unifying unsupervised graph-level anomaly detection and out-of-distribution detection: A benchmark," in *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- 5 **Xu Shen**, Pietro Lio, Lintao Yang, Ru Yuan, Yuyang Zhang, and Chengbin Peng, "Graph rewiring and preprocessing for graph neural networks based on effective resistance," *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2024.
- 6 **Xu Shen**, Yili Wang, Kaixiong Zhou, Shirui Pan, and Xin Wang, "Optimizing ood detection in molecular graphs: A novel approach with diffusion models," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD2024)*, 2024, pp. 2640–2650.

Awards

- 2025
 - **National Scholarship.**
 - **China FAW-Volkswagen Scholarship.**
- 2024
 - **Graduate Merit Scholarship (First Class)**, Jilin University.
 - **Master Academic Scholarship**, Jilin University.
- 2023
 - **Master Academic Scholarship**, Jilin University.
- 2022
 - **Zhejiang Provincial Government Scholarship**, Zhejiang Province.
- 2020
 - **Zhejiang Provincial Government Scholarship**, Zhejiang Province.

Selected Research Experiences

- **Unsupervised error detection of steps in the execution trajectories of multi-agent systems**
 - **Objective:** Utilizing a frozen parameters LLM with learnable step prototypes to achieve real-time unsupervised error detection
 - **Methods:**
 - * Utilizing the next step reconstruction capability of LLM to restructure the step in order to identify samples that deviate from the normal trajectory.
 - * Using prototypes to enhance the detection ability in cases of insufficient context.
 - **Outcome:** A research paper published in **ACL 2026** (Publication #1)
- **Define and evaluate the unfaithfulness of the chain-of-thought at the instance level**
 - **Objective:** Define and complete the annotation dataset for the unfaithfulness of instance-level CoT, using it to conduct the evaluation of existing detection methods.
 - **Methods:**
 - * Collect the CoT from four different LLMs on four different datasets and annotate them, resulting in over 300 unfaithful samples.

- * Use the labeled data to conduct a comprehensive evaluation of the existing methods, and draw insights related to the results.
- **Outcome:** A research paper published in **ICLR 2026** (Publication #2)
- **Analyze the influence of different communication topologies on LLM-based MAS**
 - **Objective:** Generate a reliable communication topology for LLM-based MAS that balances error and insight propagation
 - **Methods:**
 - * This paper analyzed the existing communication topology using causal tracing technique, and concluded that the sparsity help suppress error propagation and the density facilitate insight propagation
 - * Based on the current query, a topology that balances sparsity and density is generated to ensure robustness and efficiency.
 - **Outcome:** A research paper published in **EMNLP 2025** (Publication #3)

Service

Conference Reviewer: AAAI (2026), AISTATS (2025-2026), NeurIPS (2024-2025),
ICLR (2025-2026), ICML (2025-2026), KDD (2025-2026).